Plasma Instabilities in Heliospheric Magnetic Flux Ropes

Code accompanying the paper *Plasma Instabilities in Heliospheric Magnetic Flux Ropes: the Effect of Internal Twist and Lorentz Forces*, by Samuel Capellas Coderque, Laura Balmaceda, and Teresa Nieves-Chinchilla, submitted to Solar Physics.

The code computes growth rates of linear MHD instabilities in cylindrical flux ropes with a free boundary, for any azimuthal wavenumber m. It is applied to a modified Gold-Hoyle model, where the force parameter ζ adds Lorentz forces to the force-free case (ζ = 1), and to 15 ICMEs observed by Parker Solar Probe and Solar Orbiter.

Repo contents

File	Description
`mag_field.py`	Modified Gold-Hoyle magnetic field profiles: Bz, Bθ and their derivatives
`main_notebook.ipynb`	Stability analysis of the modified Gold-Hoyle model. Reproduces Figures 4 to 8
`realdata_notebook.ipynb`	Applies the analysis to 15 real PSP/SolO ICME observations (Section 4.3). Runs independently and produces Table 1 and Figure 9
`icme_instability_results.csv`	Output of `realdata_notebook`, numerical results per event
`icme_tables.tex`	LaTeX table generated from the above
`sheet_sigma_q.csv`	Maximum kink growth rate as a function of q for ζ = 1 (Figure 9)
`sheet_zeta1.csv`, `sheet_zeta1.5.csv`	Maximum growth rates for ζ = 1 and ζ = 1.5 (Figure 6)
`sheet_m1.csv`	Stability curves for m = 1 and three thresholds λ_0 (Figure 7, left)
`sheet_l1e-7.csv`	Stability curves for m = 1, 2, 3 at $\lambda_0 = 10^{-7}$ (Figure 7 right, and Figure 8)

Both notebooks import from `mag_field.py`, so keep it in the same directory.

Usage

For the figures, run `main_notebook.ipynb` top to bottom. The sheet CSVs are already provided, so the slow computations are bypassed automatically. The code that generates them is included but commented out; if you rerun it, the output is saved under a random file name that has to be renamed to the one the notebook loads.

The plots are rendered with LaTeX (`text.usetex = True`). If you don't have LaTeX installed, set it to `False`.

For Table 1 and Figure 9, run `realdata_notebook.ipynb` top to bottom. It produces `icme_instability_results.csv` and `icme_tables.tex`, and plots λ(k) for each event; if the red dot marking the maximum sits at the edge of the k range, widen the range. The λ(k) plots keep the event numbering of Good et al. (2023); Table 1 renumbers the events 1 to 15.

Dependencies

A standard scientific Python environment (e.g. Anaconda, Python 3.8+) will have everything needed: `numpy`, `scipy`, `pandas`, `matplotlib`, and `ipywidgets`. The figures also need a LaTeX installation (see above).

Revision (October 2026)

The force parameter has been renamed from α to ζ, the range of k in the search for the maximum growth rate has been corrected (Figure 6), and the mission data analysis (Section 4.3, Table 1, Figure 9) has been redone with a new radius estimate, $q = 2\pi|\Gamma|R$, and helium included in the mass density. Parts of the code were written with the help of the AI assistant Claude (Anthropic) and reviewed by the authors.

Citation

If you use this code, please cite the paper (reference to be added on publication). It was first developed for S. Capellas Coderque's undergraduate thesis (2022), http://hdl.handle.net/2117/371822.